

TrackB Configurable Options Reference

This document lists **all configurable elements** in TrackB that you can modify for different training and evaluation experiments.

📁 Config Files

Located in `local_extraction/configs/` :

Config File	Purpose
<code>trackB.yaml</code>	Main TrackB config (default)
<code>trackB_resnet18_baseline.yaml</code>	ResNet18 baseline preset
<code>trackB_videomae_ego.yaml</code>	VideoMAE egocentric preset
<code>trackC.yaml</code>	TrackC pruning config
<code>base.yaml</code>	Shared base settings

Usage:

```
python -m trackB.trackB_train_loader --config trackB
python -m trackB.trackB_train_loader --config trackB_videomae_ego
```

🧠 Model Architecture

Backbone / Tokenizer

Parameter	Values	Default	Description
<code>model.tokenizer.video_backbone</code>	"resnet18", "videomae_ego"	"resnet18"	Feature extraction backbone
<code>model.tokenizer.pretrained</code>	true/false	true	Use pretrained weights
<code>model.tokenizer.freeze_backbone</code>	true/false	true	Freeze backbone weights
<code>model.tokenizer.img_size</code>	int	224	Input image size
<code>model.tokenizer.time_len</code>	int	8 (resnet), 16 (videomae)	Frames in temporal window
<code>model.tokenizer.time_stride</code>	int	2	Step between frames

VideoMAE-Specific

Parameter	Values	Default	Description
<code>model.tokenizer.videomae.hf_model_name</code>	HuggingFace model name	"MCG-NJU/videomae-base"	HF pretrained model
<code>model.tokenizer.videomae.weights_path</code>	path	null	Custom weights (overrides HF)
<code>model.tokenizer.videomae.embed_dim</code>	768/1024/1280	768	Embedding dimension
<code>model.tokenizer.videomae.patch_size</code>	int	16	Spatial patch size
<code>model.tokenizer.videomae.tubelet_size</code>	int	2	Temporal patch size
<code>model.tokenizer.videomae.depth</code>	int	12	Transformer depth
<code>model.tokenizer.videomae.num_heads</code>	int	12	Attention heads
<code>model.tokenizer.videomae.freeze_encoder</code>	true/false	true	Freeze encoder

Projector

Parameter	Values	Default	Description
<code>model.projector.in_dim</code>	int	512 (resnet), 768 (videomae)	Input dimension
<code>model.projector.out_dim</code>	int	256	Output token dimension

Fusion Module (FGTP)

Parameter	Values	Default	Description
<code>model.fusion.dim</code>	int	256	Token dimension
<code>model.fusion.layers</code>	int	2	Number of fusion layers
<code>model.fusion.heads</code>	int	8	Attention heads
<code>model.fusion.dropout</code>	float	0.1	Dropout rate

Prediction Head

Parameter	Values	Default	Description
<code>model.head.dim</code>	int	256	Input dimension
<code>model.head.hidden</code>	int	256	Hidden layer size
<code>model.head.num_classes</code>	int	2	Binary next-active classes
<code>model.head.dropout</code>	float	0.1	Dropout rate

 Training Configuration

Hyperparameters

Parameter	Values	Default	Description
<code>training.epochs</code>	int	20	Number of training epochs
<code>training.batch_size</code>	int	8	Batch size
<code>training.lr</code>	float	1e-3	Learning rate
<code>training.min_lr</code>	float	1e-5	Minimum LR (for scheduler)
<code>training.warmup_epochs</code>	float	1.0	Warmup period

Regularization

Parameter	Values	Default	Description
<code>training.label_smoothing</code>	float	0.05	Label smoothing factor
<code>training.weight_decay</code>	float	0.01	Weight decay

Data

Parameter	Values	Default	Description
<code>training.candidate_limit</code>	int	16	Max candidates per frame
<code>training.normalize_ttc</code>	true/false	true	Normalize TTC to [0,1]

Checkpointing

Parameter	Values	Default	Description
<code>training.save_epoch_checkpoints</code>	true/false	true	Save each epoch
<code>training.save_best_checkpoint</code>	true/false	true	Save best model

Early Stopping

Parameter	Values	Default	Description
<code>training.early_stopping.enabled</code>	true/false	true	Enable early stopping
<code>training.early_stopping.patience</code>	int	5	Epochs without improvement
<code>training.early_stopping.monitor</code>	"mAP", "accuracy", "ttc_mae"	"mAP"	Metric to monitor
<code>training.early_stopping.mode</code>	"max", "min"	"max"	Optimization direction

Evaluation During Training

Parameter	Values	Default	Description
training.eval_every	int	1	Evaluate every N epochs

Mixed Precision

Parameter	Values	Default	Description
training.amp	true/false	false	Automatic mixed precision

🎯 Multi-Task Learning

Parameter	Values	Default	Description
multi_task.enabled	true/false	true	Enable multi-task heads
multi_task.predict_noun	true/false	true	Predict noun class
multi_task.predict_verb	true/false	true	Predict verb class
multi_task.predict_ttc	true/false	true	Predict time-to-contact
multi_task.ttc_mode	"reg" , "bin"	"reg"	TTC as regression or classification
multi_task.ttc_bins	int	5	Number of TTC bins (if bin mode)
multi_task.ttc_thresholds	list	[0.5, 1.0, 2.0]	TTC bin boundaries

Loss Weights

Parameter	Values	Default	Description
multi_task.loss_weights.next_active	float	1.0	Binary next-active loss weight
multi_task.loss_weights.noun	float	1.0	Noun classification loss weight
multi_task.loss_weights.verb	float	1.0	Verb classification loss weight
multi_task.loss_weights.ttc	float	1.0	TTC loss weight

📊 Evaluation Configuration

Basic Settings

Parameter	Values	Default	Description
evaluation.batch_size	int	8	Eval batch size
evaluation.compute_map	true/false	true	Compute mean AP
evaluation.compute_ttc_mae	true/false	true	Compute TTC MAE

Hotspot Priors (PEAR-style)

Parameter	Values	Default	Description
evaluation.hotspot_priors.enabled	true/false	true	Use hotspot priors
evaluation.hotspot_priors.path	path	local_extraction/v2/hotspot_priors_train_logodds_min10.json	Priors file
evaluation.hotspot_priors.alpha	float	0.3	Prior blending weight

CLIP Re-ranking

Parameter	Values	Default	Description
evaluation.clip_rerank.enabled	true/false	true	Use CLIP re-ranking
evaluation.clip_rerank.model	"ViT-B/32" , "ViT-L/14" , etc.	"ViT-B/32"	CLIP model variant

Parameter	Values	Default	Description
<code>evaluation.clip_rerank.weight</code>	float	<code>0.3</code>	CLIP score blending weight

Visualization

Parameter	Values	Default	Description
<code>evaluation.save_overlays</code>	<code>true/false</code>	<code>true</code>	Save visualization overlays
<code>evaluation.topk_overlay</code>	int	<code>3</code>	Number of top predictions to visualize

 Data Sources

Parameter	Values	Default	Description
<code>data.stageB_run</code>	path or <code>null</code>	<code>null</code> (auto-detect)	Stage B run directory
<code>data.train_manifest</code>	path or <code>null</code>	<code>null</code> (auto-detect)	Training manifest
<code>data.val_manifest</code>	path or <code>null</code>	<code>null</code> (auto-detect)	Validation manifest
<code>data.tokens_root</code>	path or <code>null</code>	<code>null</code>	Pre-extracted tokens directory

 TrackC: RGTP Pruning

Parameter	Values	Default	Description
<code>rgtp.enabled</code>	<code>true/false</code>	<code>true</code>	Enable RGTP pruning
<code>rgtp.rate</code>	float (0.0-0.95)	<code>0.1</code>	Fraction of tokens to prune
<code>rgtp.min_keep</code>	int	<code>2</code>	Minimum tokens to keep
<code>rgtp.temporal_decay</code>	float	<code>0.6</code>	Rollout vs motion blend factor
<code>rgtp.logit_fill</code>	float	<code>-12.0</code>	Logit value for pruned tokens

Instrumentation

Parameter	Values	Default	Description
<code>instrumentation.record_latency</code>	<code>true/false</code>	<code>true</code>	Measure inference latency
<code>instrumentation.record_vram</code>	<code>true/false</code>	<code>true</code>	Record VRAM usage
<code>instrumentation.record_flops</code>	<code>true/false</code>	<code>true</code>	Profile FLOPs

 CLI Overrides

Training script supports these CLI arguments:

```
python -m trackB.trackB_train_loader \  
  --config trackB           # Config preset  
  --epochs 30               # Override epochs  
  --batch_size 16           # Override batch size  
  --lr 0.0005               # Override learning rate  
  --demo                   # Quick demo mode
```

Evaluation script supports:

```
python -m trackB.trackB_eval \  
  --checkpoint path/to/checkpoint.pt
```

TrackC pruning:

```
python -m trackC.trackC_pruning \
  --config trackC          # Config preset
  --checkpoint path/to/checkpoint.pt
  --rgtp_rate 0.3          # Override pruning rate
  --no_pruning             # Disable pruning (baseline)
  --video_backbone videomae_ego
```

Recommended Ablation Experiments

1. Backbone Comparison

```
# ResNet18 baseline
python -m trackB.trackB_train_loader --config trackB_resnet18_baseline

# VideoMAE egocentric
python -m trackB.trackB_train_loader --config trackB_videomae_ego
```

2. Learning Rate Sweep

```
python -m trackB.trackB_train_loader --lr 0.0001
python -m trackB.trackB_train_loader --lr 0.0005
python -m trackB.trackB_train_loader --lr 0.001
python -m trackB.trackB_train_loader --lr 0.005
```

3. Multi-Task Ablation

Edit `trackB.yaml` :

```
multi_task:
  enabled: false # Disable all auxiliary tasks
```

4. Loss Weight Tuning

Edit `trackB.yaml` :

```
multi_task:
  loss_weights:
    next_active: 2.0 # Increase primary task weight
    noun: 0.5
    verb: 0.5
    ttc: 0.5
```

5. Pruning Rate Sweep (TrackC)

```
python -m trackC.trackC_pruning --rgtp_rate 0.0 # Baseline
python -m trackC.trackC_pruning --rgtp_rate 0.1
python -m trackC.trackC_pruning --rgtp_rate 0.3
python -m trackC.trackC_pruning --rgtp_rate 0.5
```

Your Best ResNet18 Config

From `trackB_best.pt` checkpoint:

```
training:
  epochs: 20
  batch_size: 8
  lr: 0.001
  min_lr: 1e-05
  warmup_epochs: 1.0
  label_smoothing: 0.05
  candidate_limit: 16
  normalize_ttc: true
  amp: false
  early_stopping:
    patience: 5
    monitor: "mAP"
    mode: "max"

multi_task:
  enabled: true
  ttc_mode: "reg" # regression
  loss_weights:
    next_active: 1.0
    noun: 1.0
    verb: 1.0
    ttc: 1.0

model:
  tokenizer:
    video_backbone: "resnet18"
```

Result: mAP 35.80%